

Lesson 01 Demo 01

Implementing DREAD Model

Objective: To implement a DREAD (Damage, Reproducibility, Exploitability, Affected users, and Discoverability) model for assessing and prioritizing security risks, enabling systematic evaluation of vulnerabilities in web applications

Tools required: Python

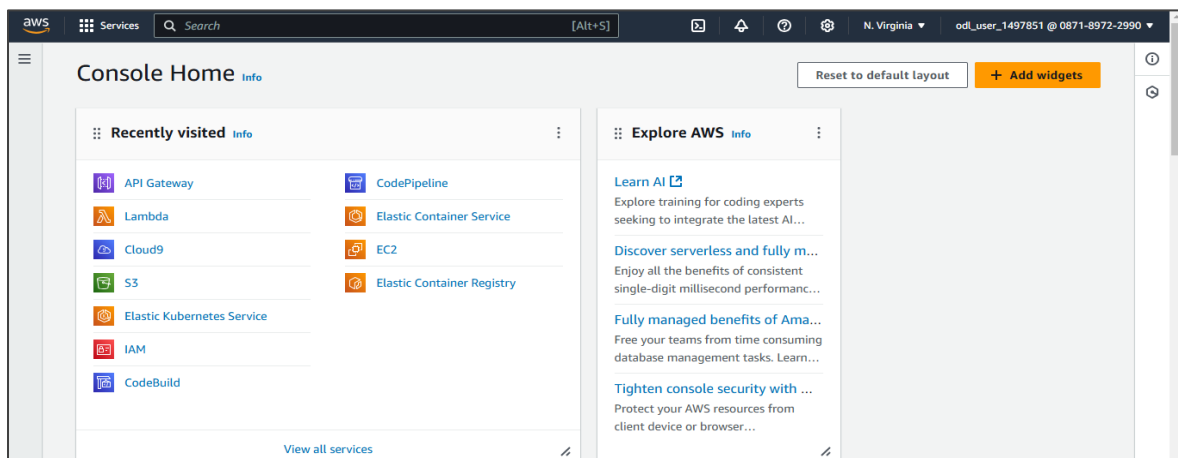
Prerequisites: None

Steps to be followed:

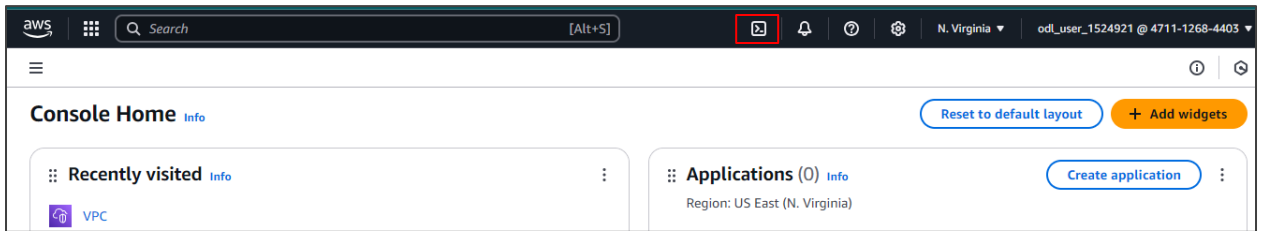
1. Implement the DREAD model using a Python script

Step 1: Implement the DREAD model using a Python script

- 1.1 Open the **AWS Management Console** in your practice environment



1.2 Click on the cloud shell icon for the terminal



1.3 Run the following command to verify if Python is installed:

python3 --version

```
[cloudshell-user@ip-10-130-76-22 ~]$ python3 --version
Python 3.9.16
[cloudshell-user@ip-10-130-76-22 ~]$
```

1.4 Execute the following command to create and navigate to the directory:

mkdir python

cd python

```
Python 3.9.16
[cloudshell-user@ip-10-130-76-22 ~]$ mkdir python
[cloudshell-user@ip-10-130-76-22 ~]$ cd python
[cloudshell-user@ip-10-130-76-22 python]$
```

1.5 Run the following command to create a file:

nano dread_model.py

```
[cloudshell-user@ip-10-130-76-22 python]$ nano dread_model.py
[cloudshell-user@ip-10-130-76-22 python]$
```

1.6 Once the file opens, add the code given below, save it, and then exit:

```
class DREADRiskModel:
    def __init__(self, damage_potential, reproducibility, exploitability, affected_users,
discoverability):
        self.damage_potential = damage_potential
        self.reproducibility = reproducibility
        self.exploitability = exploitability
        self.affected_users = affected_users
        self.discoverability = discoverability

    def calculate_risk(self):
        risk_score = (self.damage_potential + self.reproducibility +
            self.exploitability + self.affected_users +
            self.discoverability)
        return risk_score

    def print_risk_assessment(self):
        print("DREAD Risk Model Assessment:")
        print(f"Damage Potential: {self.damage_potential}")
        print(f"Reproducibility: {self.reproducibility}")
        print(f"Exploitability: {self.exploitability}")
        print(f"Affected Users: {self.affected_users}")
        print(f"Discoverability: {self.discoverability}")
        print(f"Total Risk Score: {self.calculate_risk()}")

    def get_input():
        print("Please rate each parameter from 1 (low risk) to 10 (high risk):\n")
        damage_potential = int(input("Damage Potential: "))
        reproducibility = int(input("Reproducibility: "))
        exploitability = int(input("Exploitability: "))
        affected_users = int(input("Affected Users: "))
        discoverability = int(input("Discoverability: "))
        return damage_potential, reproducibility, exploitability, affected_users,
discoverability

    def main():
        print("Welcome to the DREAD Risk Assessment Model.\n")
        damage_potential, reproducibility, exploitability, affected_users, discoverability =
get_input()
        dread_model = DREADRiskModel(damage_potential, reproducibility, exploitability,
affected_users, discoverability)
        dread_model.print_risk_assessment()

if __name__ == "__main__":
```

main()

```
us-east-1 +  
  
GNU nano 5.8 dread_model.py Modified  
print("Please rate each parameter from 1 (low risk) to 10 (high risk):\n")  
damage_potential = int(input("Damage Potential: "))  
reproducibility = int(input("Reproducibility: "))  
exploitability = int(input("Exploitability: "))  
affected_users = int(input("Affected Users: "))  
discoverability = int(input("Discoverability: "))  
return damage_potential, reproducibility, exploitability, affected_users, discoverability  
  
def main():  
    print("Welcome to the DREAD Risk Assessment Model.\n")  
    damage_potential, reproducibility, exploitability, affected_users, discoverability = get_input()  
    dread_model = DREADRiskModel(damage_potential, reproducibility, exploitability, affected_users, discoverability)  
    dread_model.print_risk_assessment()  
  
if __name__ == "__main__":  
    main()
```

Note: To save and exit the script, use CTRL + O>> press enter>> CTRL + X

1.7 Once the Python script is created, run it using the following command and follow the prompts to input ratings for the DREAD parameters as shown below:

python3 dread_model.py

```
[cloudshell-user@ip-10-130-76-22 python]$ python3 dread_model.py  
Welcome to the DREAD Risk Assessment Model.  
  
Please rate each parameter from 1 (low risk) to 10 (high risk):  
  
Damage Potential: 5  
Reproducibility: 7  
Exploitability: 5  
Affected Users: 6  
Discoverability: 3  
DREAD Risk Model Assessment:  
Damage Potential: 5  
Reproducibility: 7  
Exploitability: 5  
Affected Users: 6  
Discoverability: 3  
Total Risk Score: 26  
[cloudshell-user@ip-10-130-76-22 python]$
```

The script displays the total risk score and a detailed risk assessment.

By following these steps, you have successfully implemented a DREAD model for assessing the security risks and enabling systematic evaluation of vulnerabilities in web applications.